

Invisible Barrier: Precise Matchmaking Encryption With Keyword Search From Lattices

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Abstract—Modern cloud-based data marketplaces require secure mechanisms that enable sensitive data sharing and retrieval among mutually distrustful parties under strict confidentiality, accountability, and regulatory constraints, while relying on semi-trusted cloud infrastructures. Existing cryptographic solutions address only limited aspects of these requirements and fail to jointly support bilateral access control, efficient keyword-based retrieval, and privacy protection against information leakage in unsuccessful access attempts, particularly in the presence of post-quantum adversaries. To bridge these gaps, we propose a quantum-resistant matchmaking encryption scheme with keyword search (QKS-ME), which supports flexible keyword-based ciphertext filtering and enables on-demand decryption under bilateral access control, thereby reducing communication overhead while ensuring data authenticity. Unsuccessful searches or decryption attempts reveal no information about the plaintext, keywords, or underlying access structures to either the cloud server or unauthorized receivers. We formally prove that QKS-ME achieves indistinguishability under chosen-plaintext attacks and existential unforgeability under chosen-message attacks based on standard lattice assumptions. Experimental evaluations demonstrate that QKS-ME is computationally efficient and suitable for practical data marketplace environments.

Index Terms—Matchmaking encryption, lattice, searchable encryption, data marketplace.

I. INTRODUCTION

WITH the rapid expansion of the digital economy, cloud-based data marketplaces have emerged as a critical infrastructure for collaborative intelligence, enabling the trading and sharing of high-value data such as financial intelligence, genomic records, and cyber threat intelligence among disparate entities. Real-world platforms such as Geocento [1]

and Dawex [2] exemplify this growing paradigm. However, as data migrates from closed internal silos to open cloud-based trading environments operated by semi-trusted servers, sensitive assets are increasingly exposed to confidentiality and integrity risks [3]. A curious or compromised cloud server may infer sensitive information from stored data or access patterns, undermining both privacy guarantees and market trust. At the same time, data receivers are required to verify data provenance to satisfy accountability and regulatory requirements, such as those imposed by the General Data Protection Regulation (GDPR). These practical and legal constraints jointly highlight the urgent need for secure and trustworthy data dissemination mechanisms in cloud data marketplaces.

Various cryptographic primitives have been proposed to protect sensitive data in cloud-based environments. Identity-Based Encryption (IBE) [4], [5], [6] and Attribute-Based Encryption (ABE) [7], [8] provide reliable access control but inherently enforce unilateral authorization, making them unsuitable for data marketplaces that require mutual trust between data owners and receivers. Signcryption [9], [10] achieves confidentiality and authenticity, yet lacks expressive access structures and is confined to unilateral control. To support encrypted data retrieval, Public Key Encryption with Keyword Search (PEKS) [11], [12] and Public Key Authenticated Encryption with Keyword Search (PAEKS) [13], [14] have been widely studied; however, their reliance on complex public key management limits scalability in interconnected data marketplaces. Although integrating Searchable Encryption (SE) with IBE [15], [16] reduces key management overhead and enables selective decryption, it fails to support bilateral access control.

To address these limitations, Matchmaking Encryption (ME) and Identity-Based ME (IB-ME) [17] were introduced to enable bilateral access control, where both communicating parties specify access policies that must be mutually satisfied for successful decryption. By binding ciphertexts to the sender's identity, ME also provides data authenticity and prevents forgery. Nevertheless, existing ME schemes [18], [19], [20], [21] exhibit some limitations. Several designs decouple sender authentication from decryption to improve efficiency, which inevitably leaks sensitive information, such as the receiver's acceptance intent or the sender's identity, before decryption eligibility is determined. Moreover, current ME constructions generally lack efficient keyword search capabilities, forcing receivers to download and decrypt large volumes of ciphertexts, resulting in prohibitive communication and computational overhead. Beyond

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efficiency concerns, most existing ME schemes rely on traditional cryptographic assumptions that are vulnerable to emerging quantum attacks. Although recent lattice-based IB-ME schemes [22], [23] provide post-quantum security, they are restricted to single-bit encryption and do not support practical data retrieval. Similarly, existing quantum-resistant searchable encryption schemes [24], [25], [26] primarily focus on search functionality while overlooking expressive access control and collaborative interaction between communicating parties.

A comprehensive review of existing schemes reveals the absence of a solution that simultaneously satisfies the following design objectives:

- 1) Enforcing bilateral access control augmented with keyword-based ciphertext retrieval, enabling legitimate users to identify relevant data while ensuring data authenticity and keyword confidentiality.
- 2) Achieving communication and computational efficiency with constant-size search trapdoors.
- 3) Guaranteeing that unsuccessful decryption attempts do not leak sensitive information about the plaintext, keywords, or access structures.
- 4) Providing post-quantum security based on lattice hardness assumptions to support long-term data protection.

A. Motivation

To underscore the significance of our scheme, we elucidate our motivation through the context of sensitive medical data exchange within a data marketplace. In such settings, pharmaceutical companies exchange massive datasets ranging from genomic sequences to clinical trial results with medical institutions to accelerate drug discovery and precision medicine, where encryption is required before cloud outsourcing.

This demand is evidenced by real-world collaborations such as MELLODDY [27], a consortium of pharmaceutical companies for privacy-preserving collaboration, and commercial data exchange platforms [28], where secure data sharing among mutually distrusting parties is essential.

In this ecosystem, large-scale encrypted clinical data are outsourced to the cloud, making bulk downloading and decryption impractical to identify relevant records. Consequently, the system should enable on-demand search over encrypted clinical data, enabling users to efficiently locate relevant records, such as “targeted therapy for advanced lung cancer”, without bulk data retrieval or query leakage.

However, efficiency is not the only concern. Beyond searchability, sensitive medical data exchange strictly demands bilateral access control, as establishing such a marketplace faces a fundamental dilemma: mutual mistrust.

- For the pharmaceutical company acting as the data owner, the primary concern is the protection of high-value assets. It must enforce a rigorous access structure to ensure that its proprietary research data is accessible only by specific, certified consumers ($ID = \text{Affiliated Clinical Site}$). This strict assurance is mandatory to prevent competitive leakage and intellectual property theft, which could occur if unauthorized rivals could probe or access the data.

- Conversely, the medical institute acting as the data receiver faces the risk of fraudulent data injection. When searching for sensitive treatment protocols, the retrieved data must be cryptographically bound to an authenticated source ($ID = \text{FDA-Approved Manufacturer}$). This verification is essential to mitigate the risks of data poisoning and regulatory liability. Failing to filter out unverified sources could lead to catastrophic misdiagnoses, compromise patient safety and research integrity.

Moreover, advances in quantum computing threaten the security of traditional cryptographic schemes, necessitating the transition to post-quantum secure designs. In the burgeoning digital economy, data has evolved into a high-value strategic asset, making secure and efficient trading within data marketplaces a critical imperative. Addressing these multifaceted challenges requires a robust data sharing framework that simultaneously achieves bilateral access control, data source authentication, on-demand search, and post-quantum security.

B. Contribution

To address the discussed problems, we propose a novel scheme, termed quantum-resistant matchmaking encryption with keyword search (QKS-ME). Concretely, our main contributions are summarized as follows:

- *Bilateral Access Control and On-demand Search:* We propose a novel and flexible data sharing scheme that simultaneously achieves rigorous bilateral access control and on-demand search. Our scheme empowers both communicating parties to define access structures while supporting keyword search functionality for legitimate users. Valid access privileges are granted strictly upon the intersection of keyword match and mutual authorization. This design guarantees data authenticity and confidentiality while reducing unnecessary computational overhead for receivers.
- *Data Source Identification and Trapdoor Unlinkability:* Our scheme constructs an intrinsic source identification mechanism where the data owner embeds their specific identity information into the encryption process. This effectively guarantees data provenance, prevents data tampering or forgery, thereby safeguarding the receiver’s interests. Furthermore, our search mechanism leverages the stochastic nature of lattice-based sampling algorithms to ensure trapdoor unlinkability, thereby preventing adversaries from deducing relationships between trapdoors and keywords.
- *Preventing Information Leakage on Decryption Failure:* To optimize privacy protection and safeguard sensitive information as much as possible, our scheme ensures that unauthorized receivers remain oblivious to both the plaintext and access structures. Crucially, such receivers cannot discern whether decryption failed due to a mismatch in the sender-specified structure or the receiver-specified structure. Similarly, the cloud server is prevented from discerning the root cause of the search failure; it cannot distinguish whether the failure stems from a keyword mismatch or access privilege mismatch.

- *Post-quantum Provable Security*: To resist quantum attacks, our QKS-ME is based on lattice cryptography, a promising candidate for post-quantum security. Specifically, we provide rigorous security proofs showing that our scheme achieves ciphertext indistinguishability and trapdoor unlinkability under adaptive chosen plaintext attacks, and ensures existential unforgeability under adaptive chosen message attacks. Through extensive and comprehensive experimental analyses, we confirm the superiority and practicability of our proposed scheme.

C. Organization

Section II reviews related works. Section III introduces foundational background knowledge. Section IV presents our proposed QKS-ME scheme, detailing its system model, security definitions, and concrete construction. Section V contains correctness and security proofs. Section VI presents the comprehensive experimental evaluation and performance comparisons. Finally, Section VII concludes our work.

II. RELATED WORKS

ME, an emerging cryptographic primitive proposed by Ateiese et al. [17], was realized through a black-box construction leveraging functional encryption, signatures and zero-knowledge proof. From a practical perspective, they proposed IB-ME scheme, supporting identities as policies for matching. Francati et al. [29] later presented a new IB-ME construction without random oracles. Chen et al. [30] utilized two-level anonymous IBE structure to construct IB-ME scheme. To adapt ME to fuzzy scenarios, Wu et al. [31] proposed fuzzy IB-ME construction based on fuzzy IBE, which eliminates the reliance on functional encryption and achieves security in the standard model. Subsequent research has extended the application of IB-ME to a broader range of environments, including anonymous social network [32], edge-cloud computing [33], mobile Internet of Things [34] and so on.

Various extensions have also been proposed to enrich the features of ME. For example, Cao et al. [35] introduced deniable IB-ME that simultaneously achieves plausible deniability and non-interactive mutual authentication for anonymous messaging. Yang et al. [34] designed certificateless multi-user ME to resolve key escrow issues. To facilitate flexible one-to-many data sharing, Sun et al. [36] developed a privacy-aware and security-enhanced ME scheme, achieving efficient decryption and identity anonymity. Wu et al. [37] introduced the IB-ME scheme with a flexible mechanism for revoking decryption privileges. Xu et al. [19] extended IB-ME scheme and proposed the attribute-based matchmaking encryption. However, this scheme fails to achieve the claimed properties because malicious receivers can decrypt ciphertexts originating from senders who do not comply with their access policies. Sun et al. [21] established a bilateral fine-grained access control system that secures shared health data by enabling mutual policy enforcement between data owners and requesters, yet this scheme suffers from similar vulnerabilities in which malicious receivers can circumvent verification to decrypt data while satisfying only unilateral

access control. Crucially, all aforementioned schemes demonstrate their advantages but are rooted in traditional cryptography problems, lack of consideration for quantum attacks.

To achieve post-quantum security, efforts have been made to construct IB-ME schemes based on lattice assumptions. Belfiore et al. [38] leveraged reusable computational extractors to construct IB-ME from standard lattice assumptions. Wang et al. [39] proposed a general construction of lattice-based IB-ME from hierarchical-IBE and identity-based signature. Wang et al. [23] proposed first non-black-box IB-ME scheme from lattices, but its use of norms imposed some security limitations. Tao et al. [22] introduced a lattice-based IB-ME for Internet of Things, but their reliance on outsourcing data source identification allows the semi-trusted platform to infer receiver intentions, thereby compromising privacy. Furthermore, most existing lattice-based schemes [22], [23] are restricted to single-bit encryption. The pursuit of more expressive, privacy-preserving, multi-bit lattice-based ME schemes remains a prospective and critical area of research.

With the proliferation of cloud storage and the critical need for flexible and efficient data sharing, Boneh et al. [11] formalized PEKS primitive. To resist keyword guessing attacks, researchers have integrated authentication mechanisms into PEKS [14], [40], [41], [42], [43], [44]. Li et al. [41] proposed a public key authenticated encryption with ciphertext update and keyword search scheme, which enhances retrieval efficiency via an inverted index and constant-size trapdoors. To eliminate the reliance on fully trusted proxies, Li et al. [42] designed a proxy-free PAEKS scheme capable of secure ciphertext update. Lu et al. [43] constructed a pairing-free PAEKS scheme that achieves security against adaptively-chosen-targets adversaries for mobile applications. Luo et al. [44] proposed an efficient PEKS scheme and its authenticated extension supporting multi-keyword searches with arbitrary wildcard quantities and positions. However, PAEKS disables the ability to perform a universal search on all sender's ciphertexts using a single trapdoor. This inevitably imposes certain constraints on search flexibility. To address it, Wang et al. [16] proposed the group encryption with keyword search scheme. Additionally, the heavy burden of certificate management associated with public key encryption [11], [12], [14], [40], [41], [42], [43], [44] poses a significant challenge. To keep pace with the advancement of quantum computing, post-quantum SE has garnered attention. Behnia et al. [45] proposed the first NTRU-based PEKS scheme by fast Fourier transform. Subsequently, to eliminate certificate management overhead, lattice-based Identity-Based SE (IB-SE) schemes were introduced. By integrating SE with IBE [15], [16], [46], they also support on-demand access and conserve computational computation. Lin et al. [15] proposed a lattice-based identity-based SE scheme that supports disjunctive, conjunctive, and range queries. Zhang et al. [46] designed a lattice-based proxy-oriented IBE with keyword search scheme, allowing authorized proxies to generate searchable ciphertexts while ensuring delegation security. However, existing IB-SE schemes generally enforce unilateral access control, focusing solely on the receiver's identity. They fail to address scenarios requiring bilateral access control and mutual anonymity between

TABLE I
NOTATIONS

Notations	Definitions or Descriptions
$\mathcal{D}$	The discrete Gaussian distribution.
$\ \mathbf{T}\ $	Euclidean norm of the matrix $\mathbf{T}$.
$\tilde{\mathbf{T}}$	The result of Gram-Schmidt orthogonalization to the matrix $\mathbf{T}$.
χ	The Gaussian noise distribution.
H_1, H_2	Secure hash functions.
σ, ρ	Identity of the data sender and receiver.
snd, rcv	The target identity designated by the data receiver and sender.
ID	The general representation of the identity.
K	The globally permissible keyword set.
ω_i	The i -th keyword in the keyword set K .
$\mathbf{W}_{ID}$	The result of $H_1(ID)$.
$\mathbf{u}_{\omega_i}$	The result of $H_2(\omega_i)$.
Dr	The trapdoor generated by the data receiver.

the sender and receiver. Consequently, it is imperative to harmonize unlimited search capabilities and rigorous bilateral access control under a post-quantum framework, without compromising keyword confidentiality.

III. PRELIMINARIES

A. Notation

Let bold italic lowercase letters (e.g., $\mathbf{b}$) denote column vectors, and bold italic uppercase letters (e.g., $\mathbf{B}$) denote matrices. Table I explains the frequently used notations in our scheme.

B. Lattice

Definition 1: Let $\mathbf{B} = [\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_m] \in \mathbb{R}^{m \times m}$ be a matrix whose columns are linearly independent. The m -dimensional full-rank lattice generated by $\mathbf{B}$ is $\Lambda(\mathbf{B}) = \{\sum_{i=1}^m c_i \mathbf{b}_i, c_i \in \mathbb{Z}\}$. $\Lambda(\mathbf{B})$ is called an integer lattice if it forms a discrete subgroup of $\mathbb{Z}^m$. Denote by $\tilde{\mathbf{B}}$ the Gram-Schmidt orthogonalization of $\mathbf{B}$. Let $\|\mathbf{B}\| = \max_{1 \leq i \leq m} \|\mathbf{b}_i\|$, where $\|\cdot\|$ denotes the Euclidean norm.

Definition 2: Let q be a prime, $\mathbf{B} \in \mathbb{Z}_q^{n \times m}$ a matrix, and $\mathbf{u} \in \mathbb{Z}_q^n$ a vector. The associated q -ary integer lattices are defined as follows:

- 1) $\Lambda_q(\mathbf{B}) = \{\mathbf{e} \in \mathbb{Z}_q^m \text{ s.t. } \exists \mathbf{s} \in \mathbb{Z}_q^n, \mathbf{e} = \mathbf{B}^T \mathbf{s} \pmod{q}\}$
- 2) $\Lambda_q^\perp(\mathbf{B}) = \{\mathbf{e} \in \mathbb{Z}_q^m \text{ s.t. } \mathbf{B}\mathbf{e} = \mathbf{0} \pmod{q}\}$
- 3) $\Lambda_q^\mathbf{u}(\mathbf{B}) = \{\mathbf{e} \in \mathbb{Z}_q^m \text{ s.t. } \mathbf{B}\mathbf{e} = \mathbf{u} \pmod{q}\}$

Definition 3 (Discrete Gaussians): [47] For the center $\mathbf{q} \in \mathbb{R}^m$, a positive parameter $\sigma \in \mathbb{R}$ and a lattice $\Lambda \subset \mathbb{Z}^m$, the Gaussian function over $\mathbb{R}^m$ is defined as $\rho_{\sigma, \mathbf{q}}(\mathbf{x}) = \exp(-\pi\|\mathbf{x} - \mathbf{q}\|^2/\sigma^2)$ and $\rho_{\sigma, \mathbf{q}}(\Lambda) = \sum_{\mathbf{x} \in \Lambda} \rho_{\sigma, \mathbf{q}}(\mathbf{x})$. The discrete Gaussian distribution over Λ with parameters σ and $\mathbf{q}$ is denoted by $\mathcal{D}_{\Lambda, \sigma, \mathbf{q}}$, where $\mathcal{D}_{\Lambda, \sigma, \mathbf{q}}(\boldsymbol{\mu}) = \rho_{\sigma, \mathbf{q}}(\boldsymbol{\mu})/\rho_{\sigma, \mathbf{q}}(\Lambda)$, $\forall \boldsymbol{\mu} \in \Lambda$.

C. Trapdoor Functions

Definition 4. *TrapGen*(n, m, q) [47]: Let n, m, q be parameters such that $q \geq 3$ is an odd integer and $m \geq \lceil 6n \log q \rceil$. The algorithm outputs $(\mathbf{A}, \mathbf{T}) \in \mathbb{Z}_q^{n \times m} \times \mathbb{Z}^{m \times m}$ where $\mathbf{A}$ is statistically close to a uniform matrix, and $\mathbf{T}$ is a basis for $\Lambda_q^\perp(\mathbf{A})$ satisfying $\|\mathbf{T}\| \leq O(n \log q)$ and $\|\tilde{\mathbf{T}}\| \leq O(\sqrt{n \log q})$.

Definition 5. *SamplePre*($\mathbf{A}, \mathbf{T}_A, \delta, \mathbf{u}$) [48]: Given matrices $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$ and its basis $\mathbf{T}_A \in \mathbb{Z}_q^{m \times m}$, a vector $\mathbf{u} \in \mathbb{Z}_q^n$,

and a parameter $\delta \geq \|\tilde{\mathbf{T}}_A\| \cdot \omega(\sqrt{\log m})$, the algorithm outputs a vector $\mathbf{e} \in \mathbb{Z}_q^m$ sampled from a distribution statistically close to $\mathcal{D}_{\Lambda_q^\mathbf{u}(\mathbf{A}), \delta}$ such that $\mathbf{A}\mathbf{e} = \mathbf{u} \pmod{q}$.

Definition 6: *SampleD*($\Lambda, \mathbf{T}, \tau, \mathbf{l}$) [48]: It's a randomized nearest-plane algorithm that samples from the discrete Gaussian distribution $\mathcal{D}_{\Lambda, \tau, \mathbf{l}}$ over any lattice Λ . Given an m -dimensional lattice Λ with the basis $\mathbf{T}$, a parameter $\tau > 0$, and a center $\mathbf{l} \in \mathbb{R}^m$, the algorithm outputs a vector $\mathbf{v} \in \Lambda$ whose distribution is statistically close to $\mathcal{D}_{\Lambda, \tau, \mathbf{l}}$.

Definition 7: *NewBasisDel*($\mathbf{A}, \mathbf{R}, \mathbf{T}_A, \iota$) [49]: Given a matrix $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$, an invertible matrix $\mathbf{R}$ sampled from $\mathcal{D}_{m \times m}$, a matrix $\mathbf{T}_A \in \mathbb{Z}_q^{m \times m}$ which is a short basis of $\Lambda_q^\perp(\mathbf{A})$ and a parameter $\iota \geq \|\tilde{\mathbf{T}}\| \cdot \iota_R \sqrt{m} \omega(\log^{3/2} m)$ where $\iota_R = \sqrt{n \log q} \cdot \omega(\sqrt{\log m})$, the algorithm outputs the short basis for $\Lambda_q^\perp(\mathbf{A}\mathbf{R}^{-1})$.

Definition 8: *SampleR*(1^m) [49]: Given $n \geq 2$, $q \geq 2$ and $m \geq 2n \lceil \log q \rceil$, the algorithm outputs a $\mathbb{Z}_q^{m \times m}$ -invertible low-norm matrix $\mathbf{R} \in \mathbb{Z}^{m \times m}$ from a distribution that is statistically close to $\mathcal{D}_{m \times m}$.

Definition 9: *SampleRwithBasis*($\mathbf{A}$) [49]: Let $q \geq 2$ a prime and $m \geq 2n \lceil \log q \rceil$. Given a matrix $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$, the algorithm excutes *TrapGen* and *SamplePre*, eventually outputs a $\mathbb{Z}_q^{m \times m}$ -invertible matrix $\mathbf{R}$ sampled from $\mathcal{D}_{m \times m}$ and a basis $\mathbf{T} \in \mathbb{Z}_q^{m \times m}$ for $\Lambda_q^\perp(\mathbf{A}\mathbf{R}^{-1})$.

D. Difficult Problems

Definition 10 (Shortest Independent Vectors Problem (SIVP)): [50] Given an n -dimensional lattice $\mathcal{L}$, the Shortest Independent Vectors Problem with an approximation factor $\gamma \geq 1$ (denoted as SIVP_γ) aims to find a set of n linearly independent lattice vectors $\{\mathbf{v}_i\}_{i=1}^n \subset \mathcal{L}$ such that $\max_i \|\mathbf{v}_i\| \leq \gamma \cdot \lambda_n(\mathcal{L})$, where $\lambda_n(\mathcal{L})$ denotes the n -th successive minimum of the lattice.

Definition 11 (Small Integer Solution (SIS) Problem): [48] Given an integer q , a matrix $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$ and a real parameter β , find a non-zero vector $\mathbf{e} \in \mathbb{Z}^m$ that satisfies $\mathbf{A}\mathbf{e} = \mathbf{0} \pmod{q}$, $\|\mathbf{e}\| \leq \beta$.

For any polynomially bounded m , $\beta = \text{poly}(n)$ and any prime $q \geq \beta \cdot \omega(\sqrt{n \log n})$, solving the average-case problem $\text{SIS}_{m, q, \beta}$ is as hard as approximating the shortest independent vectors problem (SIVP) in the worst case to within a factor $\gamma = \beta \cdot \tilde{O}(\sqrt{n})$.

Definition 12 (Learning with Errors (LWE) Problem): [50] Let q be a prime, n a positive integer, and χ a Gaussian noise distribution over $\mathbb{Z}_q$. In the $(\mathbb{Z}_q, n, \chi)$ -LWE problem, oracle $\mathcal{O}$ provides samples $(\mathbf{u}, v) \in \mathbb{Z}_q^n \times \mathbb{Z}_q$ drawn from one of two distributions:

- A noisy pseudo-random sampler $\mathcal{O}_s$: Output $(\mathbf{u}, v = \langle \mathbf{u}, \mathbf{s} \rangle + x)$, where $\mathbf{s} \leftarrow \mathbb{Z}_q^n$ is fixed, $\mathbf{u} \leftarrow \mathbb{Z}_q^n$ is uniform, and $x \leftarrow \chi$.
- A truly random sampler $\mathcal{O}_\S$: Output $(\mathbf{u}, v)$ sampled uniformly from $\mathbb{Z}_q^n \times \mathbb{Z}_q$.

An algorithm $\mathcal{A}$ solves the decision-LWE problem if it can distinguish $\mathcal{O}_s$ from $\mathcal{O}_\S$ with non-negligible advantage:

$$\text{Adv}_{\mathcal{A}}^{\text{LWE}} = |\Pr[\mathcal{A}^{\mathcal{O}_s} = 1] - \Pr[\mathcal{A}^{\mathcal{O}_\S} = 1]|$$

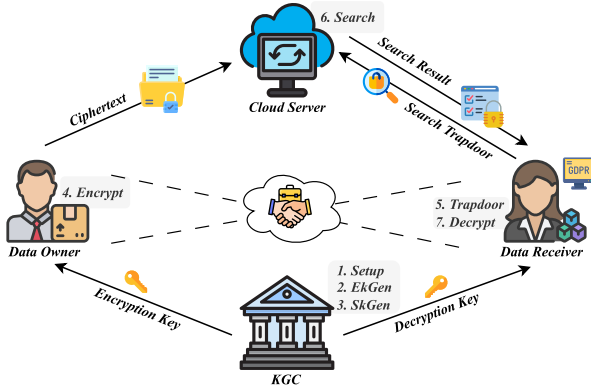


Fig. 1. System Model.

When $q = q(n)$ and $\alpha = \alpha(n) \in (0, 1)$ satisfy $\alpha q > 2\sqrt{n}$, and χ over $\mathbb{Z}_q$ is generated by $\lfloor qX \rfloor \pmod{q}$ for $X \sim \mathcal{N}(0, \alpha^2/2\pi)$, the decision-LWE problem is as hard as approximating the *shortest independent vectors problem* (SIVP) in the ℓ_2 norm to within $\tilde{O}(n/\alpha)$ in the worst case.

IV. THE CONSTRUCTION OF QKS-ME

A. System Model

As shown in Fig. 1, our proposed QKS-ME scheme comprises four entities, whose interactions are depicted therein. We detail their respective roles below.

- 1) *Key Generation Center (KGC)*: As a globally trusted authority, KGC is tasked with managing the setup of the system. It generates the system parameters and issues the secret key to each data owner and receiver.
- 2) *Data Owner*: The data owner formulates own access structure, designates the intended receiver identity, and then selects the appropriate keyword from the globally accessible keyword set based on both the data content and the primary purpose, followed by performing encryption accordingly with encryption keys obtained from KGC. The ciphertext is subsequently uploaded to cloud server to mitigate network congestion.
- 3) *Data Receiver*: The data receiver specifies the intended sender identity and retrieves the secret keys from KGC, which contain the decryption keys for on-demand match and as well the secret keys required to generate keyword-related trapdoors.
- 4) *Cloud Server (CS)*: Cloud Server, assumed to be honest but curious, functions as a backup storage and an access service node. It maintains a complete and persistent repository of ciphertexts generated by data owners, ensuring data availability. The data receiver issues a search query to CS and then CS verifies the keyword-based trapdoors over the ciphertext to find a match with the search result and sends the corresponding ciphertext components to receivers for further decryption.

B. Outline of QKS-ME

The proposed QKS-ME scheme consists of the following algorithms:

- $Setup(n) \rightarrow (PP, msk)$: The algorithm takes the security parameter n as input and outputs the public parameters PP and the main secret key msk . PP are assumed to be implicit inputs to all subsequent algorithms by default.
- $EkGen(msk, \sigma) \rightarrow ek_\sigma$: The algorithm takes the main secret key msk and the identity of owner σ as inputs and returns the encryption keys ek_σ .
- $SkGen(msk, \rho) \rightarrow sk_\rho$: The algorithm takes the main secret key msk and the identity of receiver ρ as inputs and returns the secret keys sk_ρ for both decryption and trapdoor generation.
- $Encrypt(msk, ek_\sigma, rcv, \omega_i, m) \rightarrow CT$: The algorithm takes the main secret key msk , encryption keys ek_σ , the target identity rcv designated by the sender, the appropriate keyword ω_i and information m to be encrypted as inputs and outputs the ciphertext CT .
- $Trapdoor(sk_\rho, \omega'_i) \rightarrow Dr$: The algorithm takes the secret keys sk_ρ , the target keyword ω'_i set on demand by the receiver as inputs and returns the corresponding trapdoors Dr .
- $Search(Dr, CT) \rightarrow CT^*$: The algorithm takes keyword-based trapdoors Dr , the ciphertext CT as inputs and outputs subsequent required CT^* if and only if the trapdoors Dr and CT contain the same keyword ω_i .
- $Decrypt(CT^*, snd, sk_\rho) \rightarrow m$: The algorithm takes CT^* , the target identity snd designated by the receiver and the secret keys sk_ρ as inputs and outputs m only when sk_ρ is generated based on the identity rcv specified by the sender inherent in CT , and the sender's identity exactly matches the identity specified by the receiver.

C. Security Model

Our proposed QKS-ME scheme is proven to achieve message confidentiality (indistinguishability under chosen plaintext attacks, IND-CPA) and message authenticity (existential unforgeability under chosen message attacks, EU-CMA). Notably, we formulate the IND-CPA security to encompass both ciphertext indistinguishability and trapdoor unlinkability.

The security models above are defined through interactive games between the adversary $\mathcal{A}$ and the challenger $\mathcal{C}$.

IND-CPA: $\mathcal{O}$ denotes the complete set of oracles employed in the security game, including the hash oracles $\mathcal{O}_{H_1}$, $\mathcal{O}_{H_2}$, the key oracles $\mathcal{O}_{EkGen}$, $\mathcal{O}_{SkGen}$ and the trapdoor oracle $\mathcal{O}_{Trapdoor}$. The two IND-CPA security definitions for keyword-based searchable matchmaking encryption scheme QKS-ME is specified as follows to address both cases of security requirements.

*IND-CPA Ciphertext Indistinguishability, IND-CPA-CI Game*_{QKS-ME, A}^{IND-CPA-CI}(n):

- 1) *Setup*: Upon receiving n and the maximum number of keywords k , $\mathcal{C}$ executes Setup to generate public parameters PP and main secret key msk .
- 2) *Phase 1*: $\mathcal{A}$ can query to oracles: $\mathcal{O}_{H_1}$, $\mathcal{O}_{H_2}$, $\mathcal{O}_{EkGen}$, $\mathcal{O}_{SkGen}$, $\mathcal{O}_{Trapdoor}$.
- 3) *Challenge*: $\mathcal{A}$ submits the challenge identity rcv^* along with $(M_0, M_1, \sigma_0, \sigma_1)$ to $\mathcal{C}$, where rcv^* did not appear in any query in Phase 1. Then, $\mathcal{C}$ randomly selects $b \leftarrow \{0, 1\}$.

$\{0, 1\}$ and generates the challenge ciphertext CT_b of M_b . Then, $\mathcal{C}$ sends CT_b to $\mathcal{A}$.

- 4) **Phase 2:** Repeat the procedure described in **Phase 1**.
- 5) **Guess:** $\mathcal{A}$ outputs the guess b' . $\mathcal{C}$ returns 1 if $b' = b$. Otherwise, it returns 0. The advantage of $\mathcal{A}$ in winning this game is as shown by the following expression:

$$\text{Adv}_{\text{QKS-ME}, \mathcal{A}}^{\text{IND-CPA-CI}}(n) = |\Pr[\text{Game}_{\text{QKS-ME}, \mathcal{A}}^{\text{IND-CPA-CI}}(n) = 1] - \frac{1}{2}|$$

Definition 13 (IND-CPA-CI security): Our QKS-ME achieves IND-CPA ciphertext indistinguishability if for any probabilistic polynomial-time (PPT) adversary, its advantage in winning the above game is negligible.

IND-CPA Trapdoor Unlinkability, IND-CPA-TU: Beyond data confidentiality, protecting search patterns is critical. Without trapdoor unlinkability, the server can correlate distinct queries to the same identity, enabling frequency analysis and statistical inference attacks to recover underlying keywords. QKS-ME ensures that trapdoors for the same target are unlinkable and indistinguishable from random values.

Game_{QKS-ME, A}^{IND-CPA-TU}(n):

- 1) **Setup:** Upon receiving n and the maximum number of keywords k , $\mathcal{C}$ executes Setup to generate public parameters PP and main secret key msk .
- 2) **Phase 1:** $\mathcal{A}$ issues multiple queries to oracles: $\mathcal{O}_{H_1}$, $\mathcal{O}_{H_2}$, $\mathcal{O}_{EkGen}$, $\mathcal{O}_{SkGen}$, $\mathcal{O}_{Trapdoor}$.
- 3) **Challenge:** $\mathcal{A}$ submits the challenge identity rcv^* along with keywords (ω_0, ω_1) to $\mathcal{C}$, where rcv^* did not appear in any query in Phase 1. Then, $\mathcal{C}$ randomly selects $b \leftarrow_{\$} \{0, 1\}$ and generates the challenge trapdoor $Dr_b \leftarrow \text{Trapdoor}(sk_{rcv^*}, \omega_b)$. Then, $\mathcal{C}$ sends it to $\mathcal{A}$.
- 4) **Phase 2:** Repeat the procedure described in **Phase 1**.
- 5) **Guess:** Finally, $\mathcal{A}$ outputs the guess b' . $\mathcal{C}$ returns 1 if $b' = b$. Otherwise, it returns 0. The advantage of $\mathcal{A}$ in winning this game is:

$$\text{Adv}_{\text{QKS-ME}, \mathcal{A}}^{\text{IND-CPA-TU}}(n) = |\Pr[\text{Game}_{\text{QKS-ME}, \mathcal{A}}^{\text{IND-CPA-TU}}(n) = 1] - \frac{1}{2}|$$

Definition 14 (IND-CPA-TU security): Our QKS-ME achieves IND-CPA trapdoor unlinkability if for any probabilistic polynomial-time adversary, its advantage in winning the above game is negligible.

EU-CMA: In our bilateral access control scheme, each data owner must first register with KGC to obtain encryption keys bound to their identity, and only registered data owners are able to generate valid ciphertexts that can be decrypted by receivers. Crucially, our matching mechanism allows a receiver to explicitly specify the intended sender, whereby successful decryption acts as implicit authentication of the sender's identity.

Therefore, to prevent adversaries from forging valid ciphertexts and impersonating designated senders, our scheme must guarantee existential unforgeability against chosen-message attacks. The EU-CMA game is defined as follows.

Game_{QKS-ME, A}^{EU-CMA}(n):

- 1) **Setup:** Upon receiving n and the maximum number of keywords k , $\mathcal{C}$ executes Setup to generate public parameters PP and the main secret key msk .
- 2) **Query:** $\mathcal{A}$ makes multiple polynomial queries to oracles: $\mathcal{O}_{H_1}$, $\mathcal{O}_{H_2}$, $\mathcal{O}_{EkGen}$, $\mathcal{O}_{SkGen}$, $\mathcal{O}_{Trapdoor}$.

- 3) **Forgery:** $\mathcal{A}$ forges a tuple $(C, \rho^*, \text{snd}^*)$, sends it to $\mathcal{C}$.
- 4) **Output:** $\mathcal{C}$ executes SkGen to generate dk_{ρ^*} and then executes Decrypt to recover m . $\mathcal{C}$ returns 1 if $\sigma \neq \text{snd}^*$ for any σ queried in $\mathcal{O}_{EkGen}$ and simultaneously $m \neq \perp$, then we consider that the adversary has won the game. Otherwise, it returns 0.

Definition 15 (EU-CMA security): Our QKS-ME achieves EU-CMA secure if for any probabilistic polynomial-time adversary, its advantage in winning the above game, specifically the advantage in producing a valid forgery under chosen message attacks is negligible.

D. Details of Our Scheme

- **Setup(n) $\rightarrow$ PP, msk :** Given the security parameter n , KGC accomplishes the system initialization, outputs the public parameters along with the main secret key.
 - 1) Choose the globally permissible keyword set $K = \{\omega_1, \omega_2, \dots, \omega_k\}$, suitable values for parameters n, m, q where $q = \text{poly}(n)$, $m \geq 6n \lceil \log q \rceil$ and the discrete Gaussian distribution χ . Subsequently, designate two hash functions $H_1 : \{0, 1\}^* \rightarrow \mathbb{Z}_q^{m \times m}$ whose output is an invertible matrix sampled from the distribution $D_{m \times m}$, and $H_2 : \{0, 1\}^* \rightarrow \mathbb{Z}_q^n$.
 - 2) Perform the lattice trapdoor generation algorithm $\text{TrapGen}(n, m, q)$, obtaining $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$, $\mathbf{T}_A \in \mathbb{Z}_q^{m \times m}$, where $\mathbf{T}_A$ is the basis of $\Lambda_q^\perp(\mathbf{A})$. Randomly select two vectors $\mathbf{u}_1, \mathbf{u}_2 \in \mathbb{Z}_q^n$.
 - 3) KGC outputs the public parameters $\text{PP} = \{\mathbf{A}, K, H_1, H_2, \chi\}$ and $msk = \{\mathbf{T}_A, \mathbf{u}_1, \mathbf{u}_2\}$.
- **EkGen(msk, σ) $\rightarrow$ ek_σ :** Given msk and the identity of the data owner σ , KGC generates the set of encryption key ek_σ .
 - 1) KGC randomly selects $t, g \in \mathbb{Z}_q^*$ and $\mathbf{v}_1, \mathbf{v}_2 \in \mathbb{Z}_q^{m \times m}$ such that $\mathbf{v}_1 \mathbf{v}_2 = \mathbf{V}$, where the multiplication result $\mathbf{V}$ is an invertible matrix, and KGC retains $g, \mathbf{V}^{-1} \mathbf{v}_1$. Similarly, KGC randomly chooses $\mathbf{f}_1, \mathbf{f}_2 \in \mathbb{Z}_q^{m \times m}$ ensuring $\mathbf{f}_1 \mathbf{f}_2 = \mathbf{F}$, and $\mathbf{F}^{-1} \mathbf{f}_1$ is saved.
 - 2) For the legally registered data owner, calculate

$$\mathbf{u}_1^{-1} = \left[\sum_{j=1}^n \mathbf{u}_1[j]^2 \right]^{-1} \quad \text{and} \quad \mathbf{u}_2^{-1} = \left[\sum_{j=1}^n \mathbf{u}_2[j]^2 \right]^{-1}.$$

Compute $\mathbf{e}_{\sigma 1} = t \mathbf{u}_1^{-1} \mathbf{u}_1$. To express it clearly, the result of $H_1(ID)$ is represented as $\mathbf{W}_{ID}$. KGC performs $\text{NewBasisDel}(\mathbf{A}, \mathbf{W}_{\sigma}^{-1}, \mathbf{T}_A, \iota)$ to gain $\mathbf{T}_\sigma$, executes $\text{SamplePre}(\mathbf{A} \mathbf{W}_\sigma, \mathbf{T}_\sigma, t^{-1} \mathbf{u}_2, \delta)$ sampling out $\mathbf{e}_{\sigma 2} \in \mathbb{Z}_q^m$.

- 3) KGC iteratively uses $\text{SampleD}(\Lambda_q^\perp(\mathbf{A}), \mathbf{T}_A, \tau, \mathbf{0})$ to generate the invertible matrix $\mathbf{M} \in \mathbb{Z}_q^{m \times m}$ as $\mathbf{e}_{\sigma 3}$ and then performs $\text{NewBasisDel}(\mathbf{A}, \mathbf{M}^{-1}, \mathbf{T}_A, \iota)$ to derive $\mathbf{T}_H$ which is the basis of $\Lambda_q^\perp(\mathbf{A} \mathbf{M})$, while retaining $\mathbf{A} \mathbf{M}, \mathbf{T}_H$. For brevity and clarity, $\mathbf{A} \mathbf{M}$ is denoted by $\mathbf{H}$.
- 4) KGC sends the encryption key set $ek_\sigma = \{\mathbf{v}_2, \mathbf{f}_2, g, \mathbf{e}_{\sigma 1}, \mathbf{e}_{\sigma 2}, \mathbf{e}_{\sigma 3}\}$ to the data owner.
- **SkGen(msk, ρ) $\rightarrow$ sk_ρ :** Given msk and the identity of the data receiver ρ , KGC generates the decryption keys and the

secret keys required to generate keyword-related trapdoors dk_ρ .

- 1) KGC randomly selects $o, c \in \mathbb{Z}_q^*$ and computes $d_{\rho 1} = o^{-1}u_2^{-1}u_2^T$. KGC adopts $NewBasisDel(A, W_\rho^{-1}, T_A, \iota)$ to generate T_ρ , then executes $SamplePre(AW_\rho, T_\rho, ou_1, \delta)$ sampling out $\mu \in \mathbb{Z}_q^m$. KGC utilizes the secretly saved information set and calculates $d_{\rho 2} = \mu^T V^{-1}v_1$, $d_{\rho 3} = F^{-1}f_1$ as parts of the decryption keys; $sk_1 = cg$ as one part of secret keys.
- 2) KGC iteratively invokes the gaussian sampling algorithm $SampleD(\Lambda_q^\perp(A), T_A, \tau, 0)$ to generate an invertible matrix $Q \in \mathbb{Z}_q^{m \times m}$ and computes $sk_2 = cQ^T$. Then run $NewBasisDel(H, (W_\rho Q)^{-1}, T_H, \iota)$ to obtain $T'_H \in \mathbb{Z}_q^{m \times m}$ for $\Lambda_q^\perp(HW_\rho Q)$ as sk_3 . For the sake of simplicity, we denote the multiplication result of $HW_\rho Q$ as sk_4 .
- 3) KGC sends the secret key set $sk_\rho = \{d_{\rho 1}, d_{\rho 2}, d_{\rho 3}, sk_1, sk_2, sk_3, sk_4\}$ to the user.
- $Encrypt(msk, ek_\sigma, rcv, \omega_i, m) \rightarrow CT$: Given msk , encryption keys ek_σ , the target identity rcv , the designated keyword ω_i and the plaintext $m \in \{0, 1\}^l$, the data owner generates ciphertext CT as follows.
 - 1) For this chosen keyword in K , compute $H_2(\omega_i)$ for $1 \leq i \leq k$, and represent the hashed result as $u_{\omega_i} \in \mathbb{Z}_q^n$. Calculate $u_{\omega_i}^{-1} = [\sum_{j=1}^n u_{\omega_i}[j]^2]^{-1}$.
 - 2) The data owner randomly chooses $s \in \mathbb{Z}_q^l$, $\eta, \gamma \in \mathbb{Z}_q^*$, samples a random noise vector $\chi_l \in \chi^l$ and then computes following ciphertext components:

$$\begin{aligned} C_0 &= \gamma s + m \left\lfloor \frac{q}{2} \right\rfloor + \chi_l \in \mathbb{Z}_q^l \\ C_1 &= \eta \gamma f_2 e_{\sigma 2} \in \mathbb{Z}_q^m \\ C_2 &= v_2 \eta^{-1} W_{rcv}^T A^T e_{\sigma 1} s^T \in \mathbb{Z}_q^{m \times l} \\ C_\omega &= u_{\omega_i}^{-1} W_{rcv}^T e_{\sigma 3}^T A^T u_{\omega_i} g \in \mathbb{Z}_q^m \end{aligned}$$

Eventually, the data owner generates $CT = \{C_0, C_1, C_2, C_\omega\}$ and uploads it to CS.

- $Trapdoor(sk_\rho, \omega'_i) \rightarrow Dr$: Given the target keyword ω'_i and sk_ρ , the data receiver generates keyword-based trapdoors for on-demand decryption.
 - 1) The data receiver chooses $l \in \mathbb{Z}_q^*$, executes $SamplePre(sk_4, sk_3, lu_{\omega'_i}, \delta)$ to gain $d_\omega \in \mathbb{Z}_q^m$.
 - 2) The data receiver then computes following trapdoor components: $dr_1 = d_\omega^T sk_2$, $dr_2 = lsk_1$ and sends $Dr = \{dr_1, dr_2\}$ to the cloud server to perform trapdoor verification for search.
- $Search(Dr, CT) \rightarrow CT^*$: Given the trapdoors Dr from the data receiver and the ciphertext CT , CS is responsible for executing the following deterministic polynomial-time algorithm to verify whether the data receiver's trapdoors tally with the current keyword included in CT .
 - 1) The CS executes this operation to identify ciphertexts containing the keyword ω'_i as well as the specific receiver's identity embedded in Dr .

$$dr'_2 = dr_1 C_\omega \quad (1)$$

- 2) The CS checks whether the value of dr'_2 computed in (1) matches the entry dr_2 in the trapdoor information Dr . If the data receiver is the intended recipient of the ciphertext and the inherent keyword is correctly matched, then $dr_2 = dr'_2$ will hold, and the CS returns the corresponding ciphertext components $CT^* = \{C_0, C_1, C_2\}$; otherwise, it returns 0.

- $Decrypt(CT^*, snd, sk_\rho) \rightarrow m$: Taking as inputs CT^* , the target sender's identity snd and sk_ρ , the data receiver performs as follows for decryption.

- 1) The data receiver verifies whether their identity satisfies the access structure defined by the data owner, and simultaneously checks whether the data owner's identity conforms to his or her access structure and ensures the authenticity of the message. Calculate:

$$\varepsilon_1 = d_{\rho 1} A H_1(snd) d_{\rho 3} C_1 \quad (2)$$

$$\varepsilon_2 = d_{\rho 2} C_2 \quad (3)$$

- 2) The data receiver integrates the results from the above formula (2), (3) to cover plaintext:

$$\begin{aligned} m' &= C_0 - (\varepsilon_1 \varepsilon_2)^T \\ &= m \left\lfloor \frac{q}{2} \right\rfloor + \chi_l \in \mathbb{Z}_q^l \end{aligned}$$

Compare the absolute value of $m'[\lambda]$ with $\lfloor \frac{q}{4} \rfloor$, $1 \leq \lambda \leq l$, if $|m'[\lambda]| \geq \lfloor \frac{q}{4} \rfloor$, assign $m'[\lambda] = 1$; else, assign $m'[\lambda] = 0$.

E. System Architecture

For large-scale data dissemination, our scheme can be deployed as a Key Encapsulation Mechanism (KEM) within a hybrid encryption framework. The actual data payload is encrypted once using a symmetric cipher, while only the lightweight symmetric keys are encrypted using our QKS-ME scheme for different receivers. This ensures that the storage overhead remains linear only to the number of receivers' keys, not the data size.

V. CORRECTNESS AND SECURITY ANALYSIS

A. Correctness

Theorem 1: If the keyword selected by the data receiver matches the keyword embedded in the ciphertext, and the receiver possesses valid access rights, then the trapdoor verification succeeds.

Proof: Let ω_i be the keyword contained in CT_ω and ω'_i be that in Dr . Assuming that $rcv = \rho$, $\omega_i = \omega'_i$, the corresponding trapdoor component in dr_1 satisfies $sk_4 d_\omega = lu_{\omega_i}$.

Based on it, the trapdoor verification process is successfully completed as follows:

$$\begin{aligned} dr_1 C_\omega &= d_\omega^T sk_2 u_{\omega_i}^{-1} W_{rcv}^T e_{\sigma 3}^T A^T u_{\omega_i} g \\ &= u_{\omega_i}^{-1} g c d_\omega^T Q^T W_{rcv}^T M^T A^T u_{\omega_i} \\ &= u_{\omega_i}^{-1} g c d_\omega^T (A M W_\rho Q)^T u_{\omega_i} \\ &= u_{\omega_i}^{-1} g c d_\omega^T sk_4^T u_{\omega_i} \end{aligned}$$

$$\begin{aligned}
&= u_{\omega_i}^{-1} gcl u_{\omega_i}^T u_{\omega_i} \\
&= lcg = lsk_1 \\
&= dr_2
\end{aligned}$$

where l is a random number chosen by the data receiver.

Therefore, QKS-ME satisfies correctness and consistency, the CS could be assured that CT and Dr contain the same keyword ω_i only when the verification equation $dr_1 C_\omega = dr_2$ holds. Finally, the CS responds with the corresponding encrypted message associated with the keyword and mutual access strategies to the data receiver, such that the data receiver could further decrypt it if the communication parties successfully match each other. $\square$

Theorem 2: If the data receiver's identity satisfies the requirements formulated by the data owner, and the owner's identity is successfully verified by the receiver, then the receiver can decrypt the ciphertext and recover the plaintext.

Proof: Combining the above Decrypt algorithm, we divide the correctness proof process into the following three steps:

- 1) Based on the *SamplePre* algorithm used in EkGen, we can hypothesize that once the data sender registers with KGC using their own identity σ to obtain the corresponding encryption key, the encryption key component respect to the identity σ satisfies the formula $AW_\sigma e_{\sigma 2} = t^{-1}u_2$. The data receiver is bound to authenticate the data sender's identity, which means $snd = \sigma$. Consequently, $AH_1(snd)e_{\sigma 2} = t^{-1}u_2$. This step serves to validate the authenticity of the shared data, simultaneously ensuring that the data remains untampered by the CS or any other external entities. The key point lies in the fact that any hairlike modification to the data would lead to decryption failure, thereby facilitating the identification of potential data anomalies. Considering these, the corresponding decryption is as follows:

$$\begin{aligned}
&d_{\rho 1} AH_1(snd) d_{\rho 3} C_1 \\
&= o^{-1} u_2^{-1} u_2^T AH_1(snd) F^{-1} f_1 \eta \gamma f_2 e_{\sigma 2} \\
&= o^{-1} u_2^{-1} \eta \gamma u_2^T AW_{snd} F^{-1} f_1 f_2 e_{\sigma 2} \\
&= o^{-1} u_2^{-1} \eta \gamma u_2^T AW_\sigma F^{-1} F e_{\sigma 2} \\
&= o^{-1} u_2^{-1} \eta \gamma u_2^T t^{-1} u_2 \\
&= o^{-1} \eta \gamma t^{-1}
\end{aligned}$$

Note: $F^{-1} f_1 f_2 = I$; $AW_\sigma e_{\sigma 2} = t^{-1}u_2$; $u_2^{-1} u_2^T u_2 = (\sum_{j=1}^n u_2[j]^2)^{-1} (\sum_{j=1}^n u_2[j]^2) = 1$ such that $\varepsilon_1 = o^{-1} \eta \gamma t^{-1}$, where η and γ denotes the random parameters used to conceal secrets during encryption, known solely to the data owner. Random parameters t, o are similarly chosen by the KGC, with their values known exclusively to the KGC.

Therefore, even a legitimate data receiver cannot detect any deviations based on the computed result at this decryption stage. The purpose of such design is to ensure that the data receiver that fails to decrypt during the matchmaking

encryption cannot determine the specific computational step that causes the error, i.e., the cause.

- 2) It is imperative for the data receiver to verify his or her access rights, which is represented by $rcv = \rho$. Based on the *SamplePre* algorithm used in SkGen, the secret key component respect to the identity ρ satisfies $AW_{rcv} \mu = AW_\rho \mu = ou_1$. Then this decryption process complies:

$$\begin{aligned}
&d_{\rho 2} C_2 \\
&= \mu^T V^{-1} v_1 v_2 \eta^{-1} W_{rcv}^T A^T e_{\sigma 1} s^T \\
&= \mu^T V^{-1} V \eta^{-1} W_{rcv}^T A^T e_{\sigma 1} s^T \\
&= \eta^{-1} \mu^T W_\rho^T A^T t u_1^{-1} u_1 s^T \\
&= \eta^{-1} t u_1^{-1} \mu^T W_\rho^T A^T u_1 s^T \\
&= \eta^{-1} t u_1^{-1} o u_1^T u_1 s^T \\
&= \eta^{-1} t o s^T
\end{aligned}$$

Note: $V^{-1} v_1 v_2 = I$; $\mu^T W_\rho^T A^T = o u_1^T$;

$$u_1^{-1} u_1^T u_1 = (\sum_{j=1}^n u_1[j]^2)^{-1} (\sum_{j=1}^n u_1[j]^2) = 1$$

such that $\varepsilon_2 = \eta^{-1} t o s^T$, where similarly η, s known solely to the data owner and t, o known exclusively to the KGC. The data receiver is unable to make any favorable judgements based on the results at this decryption stage. Combining Step 1 and 2, if decryption fails, the data receiver cannot determine from $\varepsilon_1, \varepsilon_2$ whether the failure is due to the receiver's identity not meeting the sender's requirements, or vice versa. It is also important to highlight that Step 1, 2 does not prioritize the order of execution.

- 3) Upon completing the initial two phases of computation, the data receiver collates the results $\varepsilon_1, \varepsilon_2$ and calculates as follows to recover the plaintext.

$$\begin{aligned}
&C_0 - (\varepsilon_1 \varepsilon_2)^T \\
&= C_0 - (o^{-1} \eta \gamma t^{-1} \eta^{-1} t o s^T)^T \\
&= \gamma s + m \left\lfloor \frac{q}{2} \right\rfloor + \chi_l - \gamma s \\
&= m \left\lfloor \frac{q}{2} \right\rfloor + \underbrace{\chi_l}_{\text{error vector}} \quad \square
\end{aligned}$$

B. Provable Security of QKS-ME

Theorem 3: For suitable parameters n, m, q , if the decision $(\mathbb{Z}_q, n, \chi)$ -LWE problem is hard, our scheme remains secure against any PPT adversary in IND-CPA-CI security game.

Proof: Assume there exists an adversary $\mathcal{A}$ that breaks IND-CPA ciphertext indistinguishability of QKS-ME with a non-negligible probability ϵ . On this basis, we can construct a challenger $\mathcal{C}$ to solve the decision $(\mathbb{Z}_q, n, \chi)$ -LWE problem with advantage $\epsilon/2$, by using $\mathcal{A}$ as the sub-routine algorithm that enables $\mathcal{C}$ to distinguish between samples generated by a pseudo-random noise sampler $\mathcal{O}_s$ and those from a truly random sampler $\mathcal{O}_\$$.

- **Setup:** First $\mathcal{C}$ has to obtain the decision-LWE samples $(u_i, v_i) \in \mathbb{Z}_q^n \times \mathbb{Z}_q$, where $i \in [1, 2m + 2 + n_k]$, n_k is the

number of available keywords in the globally permissible keyword set K . $\mathcal{C}$ intercalates $\mathbf{U} = [\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_m] \in \mathbb{Z}_q^{n \times m}$, executes $\text{SampleR}(1^m)$ to sample out $\mathbf{R}^*$ and sets $\mathbf{A} = \mathbf{U}\mathbf{R}^*$, $\text{PP} = \{\mathbf{A}, K, H_1, H_2, \chi\}$.

- **Phase 1:** $\mathcal{A}$ issues the adaptively generated queries for obtaining answers from the challenger $\mathcal{C}$. Q is used to describe the response process. $\mathcal{C}$ first randomly chooses rcv^* from the set of data receiver identities before any queries are issued.
 - $Q_{H_1(id)}$: Let $\mathcal{L}_1 = \{ID_i, \mathbf{C}_i, \mathbf{G}_i^{-1}, \mathbf{T}_i\}$ be a hash list maintained by $\mathcal{C}$. $\mathcal{C}$ considers the following two cases:
 - a) If $ID_i \neq rcv^*$: If ID_i has been queried before, then $\mathcal{C}$ returns $\mathbf{G}_i^{-1}$. Otherwise, $\mathcal{C}$ performs $\text{SampleRwithBasis}(\mathbf{A})$ to gain an invertible matrix $\mathbf{G}_i \in \mathbb{Z}_q^{m \times m}$ and the basis $\mathbf{T}_i \in \mathbb{Z}_q^{m \times m}$ for $\Lambda_q^\perp(\mathbf{C}_i = \mathbf{A}\mathbf{G}_i^{-1})$. Finally, $\mathcal{C}$ adds $(ID_i, \mathbf{C}_i, \mathbf{G}_i^{-1}, \mathbf{T}_i)$ into $\mathcal{L}_1$ and returns $\mathbf{G}_i^{-1}$.
 - b) If $ID_i = rcv^*$: If ID_i has been queried before, then $\mathcal{C}$ returns $\mathbf{G}_i^{-1}$. Otherwise, $\mathcal{C}$ generates $\mathbf{R}^* = \mathbf{G}_{rcv^*}^*$, $\mathbf{C}^* = \mathbf{A}\mathbf{R}^{*-1}$. Finally, $\mathcal{C}$ adds $(ID_i, \mathbf{C}^*, \mathbf{R}^{*-1}, -)$ into $\mathcal{L}_1$ and returns $\mathbf{R}^{*-1}$.
 - $Q_{H_2(\omega)}$: Let $\mathcal{L}_2 = \{\omega_i, \mathbf{k}_i, k_i^{-1}\}$ be a hash list maintained by $\mathcal{C}$. If ω_i has been queried before, $\mathcal{C}$ returns $\mathbf{k}_i$. Otherwise, $\mathcal{C}$ sets $\mathbf{k}_i = \mathbf{u}_{2m+2+i} \in \mathbb{Z}_q^n$ and calculate $k_i^{-1} = \frac{1}{\|\mathbf{k}_i\|^2}$ where $1 \leq i \leq n_k$, subsequently adds this record to $\mathcal{L}_2$, and eventually returns $\mathbf{k}_i$.
 - Q_{EkGen} : Upon receiving the query about ID_i , $\mathcal{C}$ retrieves the record $\{ID_i, \mathbf{C}_i, \mathbf{G}_i^{-1}, \mathbf{T}_i\}$ from $\mathcal{L}_1$. $\mathcal{C}$ sets $\mathbf{P} = [\mathbf{u}_{m+1}, \mathbf{u}_{m+2}, \dots, \mathbf{u}_{2m}] \in \mathbb{Z}_q^{n \times m}$, then computes $\mathbf{H} = \mathbf{P}\mathbf{R}^*$. Run $\text{SampleRwithBasis}(\mathbf{H})$ again to gain $(\mathbf{J}, \mathbf{T}_H)$ where $\mathbf{T}_H$ is the basis for $\Lambda_q^\perp(\mathbf{H}\mathbf{J}^{-1})$. $\mathcal{C}$ randomly chooses $\alpha, \beta \in \mathbb{Z}_q^*$, $\mathbf{b}, \mathbf{c} \in \mathbb{Z}_q^{m \times m}$ and calculates $\mathbf{u}_{2m+1}^{-1} = \frac{1}{\|\mathbf{u}_{2m+1}\|^2}$, $\mathbf{u}_{2m+2}^{-1} = \frac{1}{\|\mathbf{u}_{2m+2}\|^2}$, then executes SamplePre with input $\mathbf{T}_i, H_1(ID_i), \mathbf{u}_{2m+2}$ to generate $\mathbf{e}_i \in \mathbb{Z}_q^m$. Eventually $\mathcal{C}$ returns $(\mathbf{b}, \mathbf{c}, \beta, \alpha \mathbf{u}_{2m+1}^{-1} \mathbf{u}_{2m+1}, \mathbf{e}_i, \mathbf{T}_H)$ as encryption keys. If the generation of encryption key is successful, then $\mathcal{C}$ returns the corresponding answer.
 - Q_{SkGen} : Upon receiving the query about $ID_i \neq rcv^*$, $\mathcal{C}$ retrieves $H_1(ID_i) = \mathbf{G}_i^{-1}$ from $\mathcal{L}_1$ and executes SamplePre with input $\mathbf{A}\mathbf{G}_i^{-1}, \mathbf{T}_i, \mathbf{u}_{2m+1} \in \mathbb{Z}_q^n$ to obtain μ . Then $\mathcal{C}$ randomly selects $\gamma, \eta \in \mathbb{Z}_q^*$, $\mathbf{y}, \mathbf{z} \in \mathbb{Z}_q^{m \times m}$, compute $\mathbf{d}_1 = \gamma \mathbf{u}_{2m+2}^{-1} \mathbf{u}_{2m+2}^T$, $\mathbf{d}_2 = \mu^T \mathbf{y}$ and executes SampleD with input $\Lambda_q^\perp(\mathbf{C}_i), \mathbf{T}_i, \tau, \mathbf{0}$ to obtain $\mathbf{E}$, NewBasisDel with input $\mathbf{H}\mathbf{J}^{-1}, \mathbf{E}^{-1}\mathbf{B}_i, \mathbf{T}_H, \iota$ to gain $\mathbf{s}_3$. Eventually $\mathcal{C}$ returns $(\mathbf{d}_1, \mathbf{d}_2, \mathbf{z}, \gamma \mathbf{E}, \mathbf{s}_3, \mathbf{H}\mathbf{J}^{-1}\mathbf{B}_i^{-1}\mathbf{E})$ as secret keys.
 - $Q_{Trapdoor}$: Upon receiving the query about ω_i , $\mathcal{C}$ queries the list $\mathcal{L}_2$ to retrieve $H_2(\omega_i) = \mathbf{k}_i$ and chooses $o \in \mathbb{Z}_q^*$, executes $\text{SamplePre}(\mathbf{H}\mathbf{J}^{-1}\mathbf{B}_i^{-1}\mathbf{E}, \mathbf{s}_3, o\mathbf{k}_i, \delta)$ to gain $\mathbf{Tr}_{\omega_i} \in \mathbb{Z}_q^m$. Finally, $\mathcal{C}$ returns $(\mathbf{Tr}_{\omega_i}^T \gamma \mathbf{E}, o\eta)$.
- **Challenge:** $\mathcal{A}$ transmits the challenge message $\{m_0, m_1, \text{snd}_0, \text{snd}_1\}$ to $\mathcal{C}$ and submits an identity rcv^* as the challenge identity. It is guaranteed that rcv^* has not been queried during Phase 1. Then, $\mathcal{C}$ randomly selects

$b \leftarrow_{\$} \{0, 1\}$, extracts $\{v_i, 1 \leq i \leq 2m + 2 + n_k\}$ from the decision-LWE instances and smoothly generates the challenge ciphertext of m_b . $\mathcal{C}$ randomly chooses $\varsigma, \gamma \in \mathbb{Z}_q^*$, $\chi_b \in \chi$ and calculates $\mathbf{C}_0^* = \varsigma\gamma + m_b \lfloor \frac{q}{2} \rfloor + \chi_b$. Computes $\mathbf{C}_1^* = \gamma \mathbf{c} \mathbf{e}_i$, $\mathbf{C}_2^* = \alpha \mathbf{u}_{2m+1}^{-1} \varsigma \mathbf{b} \mathbf{v}_U \mathbf{v}_{2m+1}^T$ where $\mathbf{v}_U = (v_1, \dots, v_m)^T \in \mathbb{Z}_q^m$, $\mathbf{C}_\omega^* = k_i^{-1} \beta \mathbf{v}_P \mathbf{v}_{2m+2+i}^T$ where $\mathbf{v}_P = (v_{m+1}, \dots, v_{2m})^T \in \mathbb{Z}_q^m$. Upon completion of above operations, $\mathcal{C}$ sends $\{\mathbf{C}_0^*, \mathbf{C}_1^*, \mathbf{C}_2^*, \mathbf{C}_\omega^*\}$ to $\mathcal{A}$.

- **Phase 2:** Repeat as **Phase 1**.
- **Guess:** Finally, $\mathcal{A}$ must output its guess $b' \in \{0, 1\}$. $\mathcal{C}$ returns 1 if $b' = b$. Otherwise, it returns 0.
 - A noisy pseudo-random $\mathcal{O}_s$: In this case, according to Definition 11, notice that $\mathbf{v}_U = \mathbf{U}^T \mathbf{s} + \mathbf{x}$, $\mathbf{v}_P = \mathbf{P}^T \mathbf{s} + \mathbf{x}$ for some random noise vector $\mathbf{x} \in \mathbb{Z}_q^m$ distributed as χ^m , $\mathbf{s} \leftarrow_{\$} \mathbb{Z}_q^n$. Similarly, $\mathbf{v}_i = \mathbf{u}_i^T \mathbf{s} + \mathbf{x}$, in which the random noise $\mathbf{x} \in \mathbb{Z}_q$ is distributed as χ . Therefore:

$$\begin{aligned} \mathbf{C}_2^* &= \alpha \mathbf{u}_{2m+1}^{-1} \varsigma \mathbf{b} \mathbf{v}_U \mathbf{v}_{2m+1}^T \in \mathbb{Z}_q^* \\ &= \mathbf{b}(\mathbf{U}^T \mathbf{s} + \mathbf{x}) \alpha \mathbf{u}_{2m+1}^{-1} (\mathbf{s}^T \mathbf{u}_{2m+1} + x_{2m+1}) \varsigma \\ &= \mathbf{b}((\mathbf{A}\mathbf{R}^{*-1})^T \mathbf{s} + \mathbf{x}) \alpha \mathbf{u}_{2m+1}^{-1} (\mathbf{u}_{2m+1}^T \mathbf{s} + x_{2m+1})^T \varsigma \end{aligned}$$

This is computationally indistinguishable from the real ciphertext $\mathbf{C}_2 = \mathbf{v}_2 \eta^{-1} (\mathbf{A} H_1(rcv))^T \mathbf{e}_{\sigma 1} \varsigma$. Similarly, $\mathbf{C}_\omega^*$ formally satisfies $k_i^{-1} \beta (\mathbf{P}^T \mathbf{s} + \mathbf{x}_P) (\mathbf{u}_{2m+2+i}^T \mathbf{s} + x_{2m+2+i})^T = k_i^{-1} ((\mathbf{H}\mathbf{R}^{*-1})^T \mathbf{s} + \mathbf{x}_P) (\mathbf{s}^T \mathbf{k}_i + x_{2m+2+i}) \beta \in \mathbb{Z}_q^m$.

- A truly random $\mathcal{O}_s$: In this case, $\mathcal{A}$ doesn't have any advantage corresponding to b , v_i is uniform in $\mathbb{Z}_q$.

If $\mathcal{A}$ has a non-negligible advantage ϵ in breaking the IND-CPA-CI security of QKS-ME, then the challenger $\mathcal{C}$ will effectively distinguish the decision $(\mathbb{Z}_q, n, \chi)$ -LWE instance with a probability of $\epsilon/2$. The calculation process is as follows.

$$\begin{aligned} &\frac{1}{2} \Pr[b' = b \mid \mathcal{O} = \mathcal{O}_s] + \frac{1}{2} \Pr[b' \neq b \mid \mathcal{O} = \mathcal{O}_s] - \frac{1}{2} \\ &= \frac{1}{2} \left(\frac{1}{2} + \epsilon \right) + \frac{1}{2} \left(\frac{1}{2} \right) - \frac{1}{2} = \frac{\epsilon}{2} \end{aligned}$$

A security reduction establishes the existence of a probabilistic polynomial-time algorithm $\mathcal{C}$, which solves the decision $(\mathbb{Z}_q, n, \chi)$ -LWE problem with a success probability of $\frac{\epsilon}{2}$. Considering the computational hardness of the decision-LWE problem, the probability of adversary's success is computationally negligible. $\square$

Theorem 4: For suitable parameters n, m, q , if the decision $(\mathbb{Z}_q, n, \chi)$ -LWE problem is hard, our scheme remains secure against any PPT adversary in IND-CPA-TU security game.

Proof: Assume there exists an adversary $\mathcal{A}$ that breaks IND-CPA trapdoor unlinkability of QKS-ME with the non-negligible probability. Then we can construct the challenger $\mathcal{C}$ to solve the decision $(\mathbb{Z}_q, n, \chi)$ -LWE problem.

- **Setup:** First $\mathcal{C}$ has to obtain the decision-LWE samples $(\mathbf{u}_i, v_i) \in \mathbb{Z}_q^n \times \mathbb{Z}_q$, where $i \in [1, 2m + 2 + n_k]$, n_k is the

number of available keywords in the globally permissible keyword set K . $\mathcal{C}$ intercalates $\mathbf{U} = [\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_m] \in \mathbb{Z}_q^{n \times m}$, executes $\text{SampleR}(1^m)$ to sample out $\mathbf{R}^*$ and sets $\mathbf{A} = \mathbf{U}\mathbf{R}^*$, $\text{PP} = \{\mathbf{A}, K, H_1, H_2, \chi\}$.

- **Phase 1:** $\mathcal{A}$ can issue the adaptively generated queries for obtaining answers from the challenger $\mathcal{C}$. Q is used to describe the response process. The queries including hash queries, key queries and trapdoor queries are the same as Theorem 3.

- **Challenge:** $\mathcal{A}$ transmits the challenge keyword $\{\omega_0, \omega_1\}$ to $\mathcal{C}$ and along with the challenge identity rcv^* , where rcv^* has not been queried during Phase 1. Then, $\mathcal{C}$ randomly selects $b \leftarrow_{\$} \{0, 1\}$, generates the challenge trapdoor $Dr_b = (\mathbf{Tr}_{\omega_b}^T \gamma \mathbf{E}, o\eta)$.

Specifically, when generating Dr_b , $\mathcal{C}$ first chooses the random number $o \leftarrow_{\$} \mathbb{Z}_q^*$ for each query in Q_{Trapdoor} and retrieves $H_2(\omega_i) = \mathbf{k}_i$ in $\mathcal{L}_2$. Subsequently, $\mathcal{C}$ invokes the algorithms SamplePre to obtain the vector $\mathbf{Tr}_{\omega_i}$. In addition to this, $\mathcal{C}$ executes Q_{SkGen} to gain $\mathbf{sk}_2 = \gamma \mathbf{E}$ where $\gamma \leftarrow_{\$} \mathbb{Z}_q^*$ and $\mathbf{E}$ is sampled by SampleD .

As described in Section 4.2 of [48], SampleD is a randomized nearest-plane algorithm. Unlike deterministic approaches, it employs internal randomness to sample coefficients from a discrete Gaussian distribution. Since SamplePre internally invokes SampleD as a subroutine to generate the Gaussian perturbation in Section 5.3 of [48], it inherits this intrinsic stochastic property. Consequently, for any fixed input basis and center, both algorithms output distinct vectors that are statistically independent and identically distributed over the lattice.

Notably, $\mathbf{E} \leftarrow \text{SampleD}$, $\mathbf{Tr}_{\omega_b} \leftarrow \text{SamplePre}$, and any other parameters $o, \gamma, \eta \leftarrow_{\$} \mathbb{Z}_q^*$. Due to the intrinsic randomization of the discrete Gaussian sampling, the output vectors $\mathbf{E}$ and $\mathbf{Tr}_{\omega_b}$ are distinct and statistically independent across varying invocations, even when the underlying query inputs (ω, ID) are identical.

Therefore, under the property of Gaussian sampling, the component distribution of Dr_b is computationally indistinguishable from uniform over $\mathbb{Z}_q^{1 \times m} \times \mathbb{Z}_q^*$. This ensures that the adversary cannot link Dr_b to any previous queries, thereby achieving trapdoor unlinkability.

- **Phase 2:** Repeat as **Phase 1**.
- **Guess:** $\mathcal{A}$ outputs its guess $b' \in \{0, 1\}$. $\mathcal{C}$ returns 1 if $b' = b$. Otherwise, it returns 0.

Considering the computational hardness of the decision-LWE problem and the intrinsic randomization properties of the lattice sampling algorithms, the probability of adversary's success is computationally negligible. $\square$

Theorem 5: If the SIS problem is hard, our scheme remains secure against any PPT adversary in EU-CMA security games.

Proof: Assume that there exists an adversary $\mathcal{A}$ managing to forge a valid ciphertext who breaks the EU-CMA security, we can devise an algorithm $\mathcal{C}$ to solve the SIS problem. Reduction $\mathcal{C}$ is invoked on a random instance of the SIS assumption, i.e., $\mathbf{A}^* = [\mathbf{A}'\mathbf{W}'|\mathbf{u}'] \in \mathbb{Z}_q^{n \times (m+1)}$, and is asked to return an solution $\mathbf{e} \in \mathbb{Z}^{m+1}$, such that $\mathbf{A}^*\mathbf{e} = \mathbf{0} \pmod{q}$ and $\|\mathbf{e}\| \leq \beta$.

- **Setup:** $\mathcal{C}$ receives the security parameter n and the global permissible keyword set K , then generates $\mathbf{A}, H_1, H_2$, where $\mathbf{A} = \mathbf{A}'$ and H_1, H_2 are random oracles controlled by $\mathcal{C}$.
- **Phase 1:** $\mathcal{A}$ is able to submit subsequent adaptive queries. $\mathcal{C}$ first initializes two empty lists: $\mathcal{L}_1, \mathcal{L}_2$ to record responses related to the random oracles, and then responds accordingly to the following queries:
 - **H_1 Queries:** $\mathcal{C}$ holds $\mathcal{L}_1 = \{ID_i, \mathbf{B}_i^{-1}, \mathbf{T}_i, \mathbf{u}_i, \mathbf{e}_i\}$. If ID_i has been queried before, $\mathcal{C}$ returns $\mathbf{B}_i^{-1}$. Otherwise, when $ID_i \neq \text{snd}^*$, $\mathcal{C}$ generate $(\mathbf{T}_i, \mathbf{B}_i)$ by performing $\text{SampleRwithBasis}(\mathbf{A}')$ such that $\mathbf{B}_i$ is invertible and $\mathbf{T}_i$ is the basis for $\Lambda_q^\perp(\mathbf{A}'\mathbf{B}_i^{-1})$. And then randomly select $\mathbf{e}_i \in \mathbb{Z}_q^m$ computes $\mathbf{u}_i = \mathbf{A}'\mathbf{B}_i^{-1}\mathbf{e}_i$, and eventually adds this record to $\mathcal{L}_1$. When $ID_i = \text{snd}^*$, $\mathcal{C}$ selects $\mathbf{W}' = \mathbf{B}_i^{-1}$, $\mathbf{u}' = \mathbf{u}_i$ and returns $\{ID_i, \mathbf{W}', \mathbf{u}', -\}$.
 - **H_2 Queries:** Let $\mathcal{L}_2 = \{\omega_i, \mathbf{k}_i\}$ be a hash list maintained by $\mathcal{C}$, and the initial is empty. If ω_i has been queried before, then $\mathcal{C}$ returns $\mathbf{k}_i$. Otherwise, $\mathcal{C}$ chooses a random vector $\mathbf{k}_i$. Finally, $\mathcal{C}$ adds $(\omega_i, \mathbf{k}_i)$ into $\mathcal{L}_2$ and returns $\mathbf{k}_i$.
 - **$EkGen$ Queries:** Upon receiving the query about ID_i , $\mathcal{C}$ queries the list $\mathcal{L}_1$ to retrieve $H_1(ID_i) = \mathbf{B}_i^{-1}$ and the homologous $(\mathbf{u}_i, \mathbf{e}_i)$ where $\mathbf{u}_i = \mathbf{A}'\mathbf{B}_i^{-1}\mathbf{e}_i$. $\mathcal{C}$ randomly chooses $\alpha, \beta \in \mathbb{Z}_q^*$, $\mathbf{a}_i \in \mathbb{Z}_q^n$, $\mathbf{b}, \mathbf{c} \in \mathbb{Z}_q^{m \times m}$, then invokes TrapGen to sample out $\mathbf{H}, \mathbf{A}_H$. Calculate $\mathbf{a}_i^{-1} = \frac{1}{\|\mathbf{a}_i\|^2}$, $\mathbf{u}_i^{-1} = \frac{1}{\|\mathbf{u}_i\|^2}$. Eventually $\mathcal{C}$ returns $(\mathbf{b}, \mathbf{c}, \beta, \alpha\mathbf{a}_i^{-1}\mathbf{a}_i, \mathbf{e}_i, \mathbf{H})$ as encryption keys. If the generation of encryption key is successful, then $\mathcal{C}$ returns the corresponding answer. Otherwise $\mathcal{C}$ aborts (and returns a random bit).
 - **$SkGen$ Queries:** Upon receiving the query about ID_i , $\mathcal{C}$ queries the list $\mathcal{L}_1$ to retrieve $H_1(ID_i) = \mathbf{B}_i^{-1}$ and generates the secret part $\mathbf{w}$ by executing $\text{SamplePre}(\mathbf{A}'\mathbf{B}_i^{-1}, \mathbf{T}_i, \mathbf{a}_i, \delta)$. Then $\mathcal{C}$ randomly selects $\gamma \in \mathbb{Z}_q^*$, $\mathbf{y}, \mathbf{z} \in \mathbb{Z}_q^{m \times m}$, compute $\mathbf{d}_1 = \gamma\mathbf{u}_i^{-1}\mathbf{u}_i^T$, $\mathbf{d}_2 = \mathbf{w}^T\mathbf{y}$ and executes SampleD with input $\Lambda_q^\perp(\mathbf{A}'\mathbf{B}_i^{-1})$, $\mathbf{T}_i, \tau, \mathbf{0}$ to obtain $\mathbf{E} \in \mathbb{Z}_q^{m \times m}$, NewBasisDel with input $\mathbf{H}, \mathbf{E}^{-1}\mathbf{B}_i, \mathbf{A}_H, \iota$ to gain $\mathbf{s}_3$. Eventually $\mathcal{C}$ returns $(\mathbf{d}_1, \mathbf{d}_2, \mathbf{z}, \eta, \gamma\mathbf{E}, \mathbf{s}_3, \mathbf{H}\mathbf{B}_i^{-1}\mathbf{E})$ as secret keys.
 - **Trapdoor Queries:** Upon receiving the query about ω_i , $\mathcal{C}$ queries the list $\mathcal{L}_2$ to retrieve $H_2(\omega_i) = \mathbf{k}_i$ and chooses $o \in \mathbb{Z}_q^*$, executes $\text{SamplePre}(\mathbf{H}\mathbf{B}_i^{-1}\mathbf{E}, \mathbf{s}_3, o\mathbf{k}_i, \delta)$ to gain $\mathbf{Tr}_{\omega_i} \in \mathbb{Z}_q^m$. Finally, $\mathcal{C}$ returns $(\mathbf{Tr}_{\omega_i}^T \gamma \mathbf{E}, o\eta)$ as trapdoors.
- **Forgery:** $\mathcal{A}$ delivers a forgery $(CT^*, \text{snd}^*, \rho^*)$ to $\mathcal{C}$. If snd^* has been queried in Phase 1, stop the simulation and output failure. And for this forged ciphertext to be successfully decrypted, the construction must employ a proper encryption key $\mathbf{e}_i^* \in \Lambda_q^{\mathbf{u}'}(\mathbf{A}'\mathbf{W}')$. Furthermore, the encryption key for snd^* has never been queried. We analyze that to successfully pass the first important part of the decryption steps $\varepsilon_1 = \mathbf{d}_{\rho 1}\mathbf{A}H_1(\text{snd})\mathbf{d}_{\rho 3}\mathbf{C}_1$, whose calculation process requires that the equation $\mathbf{A}'H_1(\text{snd}^*)\mathbf{e}_i^* = \mathbf{u}'$ must hold, essentially $\mathcal{C}$ needs to recover $\mathbf{e}_i^* \in \mathbb{Z}_q^m$ with knowledge of

TABLE II
STORAGE COST COMPARISONS

Scheme	Ek size	Sk size	Ciphertext size	Trapdoor size
[23]	$m \log q$	$2m \log q$	$(2m + 1 + \pi) \mathcal{M} \log q$	/
[39]	$(m + n \lceil \log q \rceil)^2 \log q$	$16m^2 \log q$	$(3m + 1)(m + n \lceil \log q \rceil + \mathcal{M}) \log q$	/
[15]	/	$m^2 \log q$	$(2m + 1) \log q$	$(2m + 1) \log q$
[12]	/	$m^2 \log q$	$(m + l(m + 1)) \log q$	$m \log q$
[13]	$(4mn + m^2) \log q$	$(4mn + m^2) \log q$	$((2d + 8)ml_S) \log q$	$((2d + 8)ml_R) \log q$
Ours	$(m + n + 3m^2 + 1) \log q$	$(m + n + 4m^2 + 1) \log q$	$(2m + (m + 1)l) \log q$	$(m + 1) \log q$

π : the proof generated by the NIZK proof scheme in [23]; $|\mathcal{M}|$: the plaintext space; d : the number of time periods;
 l : the length of encrypted binary string; l_S, l_R : the number of columns of the computation-related matrices at the sender and receiver in [13];

u' and $A'W'$ achieving $A'W'e_i^* = u'$. Hence:

$$A^* \begin{bmatrix} e_{snd}^* \\ -1 \end{bmatrix} = 0$$

Thus, e that solves the SIS instance satisfies:

$$\|e\| = \left\| \begin{bmatrix} e_{snd}^* \\ -1 \end{bmatrix} \right\| \leq \sqrt{\delta^2 m + 1}$$

If this operation can be done by $\mathcal{C}$ in polynomial time, $\mathcal{C}$ will solve the SIS $_{q,m+1,\beta}$ problem with non-negligible advantage, where $\beta = \sqrt{\delta^2 m + 1}$. Considering for the computational hardness of SIS problem, the probability of adversary's success is computationally negligible. $\square$

VI. PERFORMANCE EVALUATION

To evaluate the effectiveness and computational overhead of our scheme in practical settings, we conduct simulation experiments. The Key Generation Center and Cloud Server run on Ubuntu 22.04.3 LTS with an Intel Xeon E5-2680 v4 @ 2.40 GHz CPU featuring 56 cores, 125 GB RAM, and an NVIDIA GeForce RTX 4090 GPU with 49 GB VRAM, while the Data Owner and Data Receiver operate on Windows 11 with an Intel Core i5-12600KF @ 3.70 GHz CPU and 16 GB RAM. The software environment includes Python 3.11.10, SageMath 9.5, and NumPy 2.1.2. For each configuration, ten independent trials are conducted after a warm-up run, and the reported results are averaged. The system parameters are configured as follows: $n = 256$, $q = 1031$, and $m = 6n \lceil \log q \rceil$.

A. Storage and Communication Cost

We begin by theoretically comparing our QKS-ME scheme with five schemes in terms of storage and communication overhead. To ensure a fair comparison, all schemes included in the evaluation are constructed based on lattice cryptography. As shown in Table II, our scheme achieves a compact trapdoor size, which facilitates efficient data retrieval and reduces communication bandwidth when compared with [13], [15]. Compared with [12], our scheme attains comparable trapdoor storage efficiency while enabling simplified key management by eliminating the need for certificates. Although the scheme in [23] features smaller key sizes, it does not support on-demand search functionality, which is a core capability of our construction. In addition, while schemes [13], [39] exhibit key sizes comparable

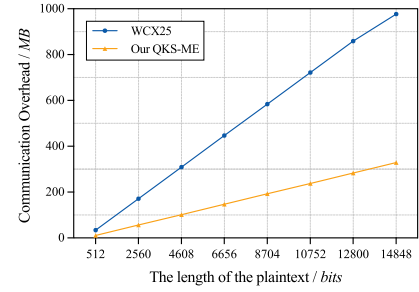


Fig. 2. Ciphertext overheads.

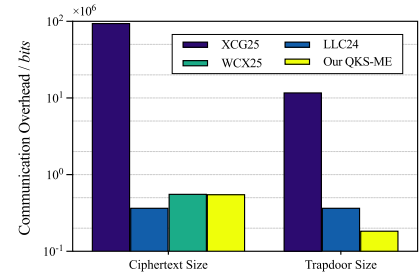


Fig. 3. Communication overhead comparison.

to ours, our scheme provides enhanced functional expressiveness. Specifically, scheme [39] does not support keyword-based search. Scheme [13], despite having a similar storage footprint and authentication function, suffers from inflexible access structures, and a restrictive trapdoor search mechanism. Overall, our scheme achieves a favorable trade-off between storage efficiency and functionality.

Table III compares the security properties and functional capabilities between our scheme and related works. As shown in Table III, none of the existing schemes provide a comprehensive feature set. For example, schemes [12], [13], [15], [41], [43] lack the capability of bilateral access control, while [23] and [39] do not support keyword search. Furthermore, several schemes lack quantum resistance and identity-based setting, whereas our scheme provides both.

To quantify communication efficiency, we evaluate the communication overhead in terms of ciphertext size and trapdoor size, as illustrated in Figs. 2 and 3. We compare our QKS-ME scheme with the XCG25 [13], LLC24 [15], and WCX25 [23]

TABLE III
FEATURE COMPARISONS

Scheme	Identity-based setting	Bilateral access control	Keyword search	Quantum resistance
[12]	×	×	✓	✓
[13]	×	×	✓	✓
[15]	✓	×	✓	✓
[23]	✓	✓	×	✓
[39]	✓	✓	×	✓
[41]	×	×	✓	×
[43]	×	×	✓	×
Ours	✓	✓	✓	✓

✓: This scheme satisfies the corresponding property; ×: This scheme dissatisfies the corresponding property.

schemes. XCG25 [13] constructs a lattice-based multi-user authenticated searchable encryption. LLC24 [15] features a post-quantum secure identity-based searchable encryption supporting disjunctive, conjunctive, and range queries. WCX25 [23] presents an identity-based ME based on lattice assumptions and zero-knowledge proof. In our simulations, the plaintext length scales in multiples of 512 bits. Note that for index-only searchable encryption schemes [13], [15], the communication overhead is irrespective of the plaintext size. In contrast, our construction is a more comprehensive matchmaking encryption scheme with keyword search that integrates bilateral access control with on-demand search. Consequently, Fig. 2 compares our scheme with an IB-ME scheme [23] by measuring ciphertext sizes across different plaintext scales, thereby providing a comprehensive view of the practical communication overhead. As shown in Fig. 2, QKS-ME achieves a shorter ciphertext length and consequently lower communication overhead by eliminating the need for the zero-knowledge proof system used in WCX25 scheme [23], while still supporting the same bilateral authentication functionality. Fig. 3 lists the communication overhead in detail. It is evident that the trapdoor size of our QKS-ME is much less than that of other schemes. Although our scheme exhibits a slightly larger ciphertext size compared to LLC24 scheme [15], this is a deliberate trade-off to accommodate bilateral access control functionality, which is absent in [15].

In summary, our scheme achieves a compact trapdoor size while maintaining an acceptable ciphertext size, thereby supporting enhanced functionality, making it a highly competitive solution for secure data marketplaces.

B. Computation Cost

To evaluate the computational efficiency of QKS-ME, we executed a series of experiments against several related constructions [13], [15], [23]. All tests were conducted under a unified experimental framework with strictly controlled variables to ensure a fair and consistent comparison. Our performance metrics include the processing time for encryption, decryption, trapdoor generation, and search.

- 1) We compare the average time cost of the Encryption, Decryption algorithms in our QKS-ME scheme with those of XCG25 scheme [13], LLC24 scheme [15] and WCX25 scheme [23]. Fig. 4 illustrates the encryption time consumption of our QKS-ME scheme in comparison with [13], [15], [23] across different security parameter

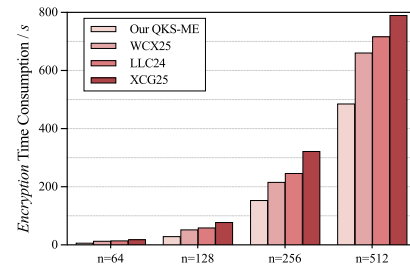


Fig. 4. Encryption cost.

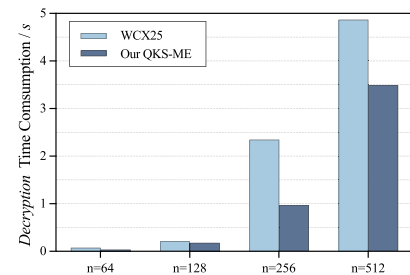


Fig. 5. Decryption cost.

scales. The empirical results indicate that as the parameter n increases, the encryption latency for all evaluated schemes exhibits a consistent upward trend, which aligns with the theoretical complexity of high-dimensional lattice operations. Notably, our QKS-ME scheme yields the most competitive computational profile in all test scenarios. As n scales to 512, the encryption overhead of QKS-ME remains under 500 seconds, whereas the competing schemes require significantly more time. Our QKS-ME scheme significantly reduces the computational burden by obviating the need for high-dimensional matrix inversion, complex Gaussian sampling, and the generation of zero-knowledge proofs during the encryption phase. As shown in Fig. 5, our QKS-ME scheme consistently achieves lower decryption latency than [23] across all parameter scales, demonstrating better scalability and computational efficiency.

- 2) We evaluate the computational cost of trapdoor generation and the search process for the proposed construction, and conduct a comparative analysis against state-of-the-art lattice-based searchable encryption schemes [13], [15].

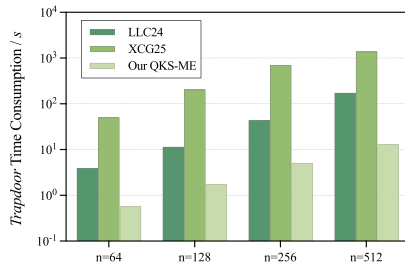


Fig. 6. Trapdoor cost.

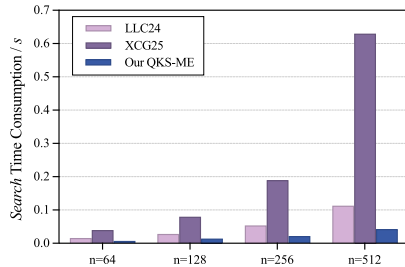


Fig. 7. Search cost.

As illustrated in Figs. 6 and 7, the proposed construction demonstrates significant advantages in keyword search operations, consistently outperforming related lattice-based schemes across various parameter scales. Fig. 6 illustrates the comparison of trapdoor generation time overhead among different schemes as the parameter n varies. As observed, the time overhead for all schemes increases as n grows from 64 to 512, which is expected due to the increased computational complexity. However, our QKS-ME scheme consistently achieves the lowest time consumption across all tested values of n . In particular, the efficiency gain in the search phase stems from our simplified computational structure. Unlike the LLC24 and XCG25 schemes, QKS-ME reduces the search process to a single vector multiplication, thereby avoiding the latency accumulation caused by multiple inner-product computations over concatenated ciphertext components and subsequent threshold comparisons. As a result, QKS-ME sustains millisecond-level response times even when the security parameter increases to 512.

VII. CONCLUSION

In this paper, we proposed QKS-ME, a novel quantum-resistant matchmaking encryption scheme with keyword search. The proposed scheme simultaneously supports bilateral access control and on-demand decryption, enabling keyword search prior to decryption while ensuring that access privileges are granted strictly upon mutual authorization. As a result, QKS-ME achieves data authenticity and privacy guarantees while significantly improving decryption efficiency. Built upon standard lattice assumptions, QKS-ME is rigorously proven to satisfy IND-CPA and EU-CMA security, while also providing trapdoor unlinkability. Furthermore, the scheme effectively hides the

underlying causes of decryption or search failures, preventing sensitive information leakage to unauthorized parties. Extensive experimental evaluations demonstrate that QKS-ME is computationally efficient, making it practical for real-world secure data-sharing applications.

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